Personal Desktop Computer Electricity Consumption:

A Full-Year Exploratory Data Analysis of Computing Behavior

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Data collection period: 13 March 2025 to 13 March 2026

Abstract

This study presents a comprehensive exploratory data analysis (EDA) of daily electricity consumption recorded by a smart plug energy monitor connected exclusively to a personal desktop computer workstation. The dataset spans one full calendar year and contains 322 recorded days, annotated with a detailed university attendance log that enables behavioral segmentation into Home Days (n = 222) and University Days (n = 100). Periods without recorded measurements correspond to intentional travel intervals rather than missing or corrupted data. The analysis reveals a strong behavioral signal linking physical work location to computing energy consumption: Home Days average 1.958 kWh of daily electricity usage compared with 0.595 kWh on University Days, representing a 229% increase. Distributional analysis shows a clear bimodal structure in the dataset, with near-complete separation between the two behavioral regimes. Unsupervised clustering using K-Means further identifies three distinct usage states corresponding to idle (~0.60 kWh), moderate (~1.80 kWh), and heavy (~2.32 kWh) computing activity. Temporal analysis across monthly, weekly, and calendar visualizations confirms that university attendance consistently suppresses desktop power consumption regardless of season. These results demonstrate that high-resolution device-level electricity measurements from a single smart plug can function as a reliable behavioral proxy, enabling inference of work location and computing activity patterns from energy data alone.

Keywords: electricity consumption, PC energy monitoring, behavioral analytics, exploratory data analysis, home office, university attendance, K-Means clustering, time series analysis.

1. Introduction

The rapid growth of remote and hybrid work arrangements has renewed interest in quantifying personal computing energy consumption. While institutional-scale energy audits are well

documented, individual-level monitoring of personal computing systems remains relatively rare. Understanding how daily behavioral choices-specifically whether a researcher works from home or commutes to an institutional workplace-translate into measurable differences in electricity consumption has both practical and environmental significance.

Previous work on computing-related behavioral monitoring has focused primarily on mobile devices, institutional infrastructure, or small-scale observational studies[1]. Digital wellbeing research has largely emphasized smartphone-based monitoring systems, while desktop and laptop computing environments remain comparatively underexplored[2]. For example, small cohort studies tracking computer usage among graduate students have demonstrated significant discrepancies between self-reported and measured activity, highlighting the value of automated monitoring systems[3]. However, such studies typically involve limited sample sizes and relatively short observation periods[3]. More broadly, personal informatics research has noted the practical challenges of sustained, high-resolution tracking of individual technology usage over long durations[4].

This study exploits an opportunistic natural experiment: a PhD researcher with a dedicated smart plug energy monitor on their desktop computer system, combined with a detailed log of university attendance dates spanning a full calendar year. The research questions addressed are: (1) How strongly does home versus university presence influence daily PC electricity consumption? (2) Does the consumption data exhibit a bimodal distribution consistent with two operating regimes? (3) Can K-Means clustering reliably identify distinct behavioral usage states from the raw energy signal alone? (4) What are the seasonal and weekday patterns in computing behavior over the year?

2. Dataset Description

2.1 Measurement Setup

A smart plug energy monitor was installed on a single power strip supplying a personal desktop computer system. The system comprises a desktop PC chassis (CPU, GPU, motherboard, storage), a monitor, speakers and associated peripherals connected to the same strip. No other household appliances share this circuit. Consequently, the daily kWh readings reflect only computing activity and system standby power.

2.2 Temporal Coverage

The dataset spans 322 recorded days between 13 March 2025 and 13 March 2026. Three contiguous gaps represent travel events during which the researcher was physically absent and the PC was unused:

Table 1. Travel gaps in the dataset (intentional absences, not imputed).

Period	Dates	Duration	Description
Travel interval 1	1 Apr – 30 Apr 2025	30 days	Research visit (Shizuoka, Japan)
Travel interval 2	8 Oct – 11 Oct 2025	4 days	Conference travel (Mumbai, India)
Travel interval 3	22 Nov – 1 Dec 2025	10 days	Conference travel (Jeju, South Korea)

These gaps are treated as behaviorally meaningful events rather than data quality issues. The first travel interval in particular - a full calendar month - provides a natural baseline: a 30-day period of zero consumption against which the rest of the year can be contrasted.

2.3 Day-Type Labeling

Each recorded day was labeled using a binary classification scheme based on an independent university-visit log. Days appearing in the visit log are labeled **University Day**; all other recorded days are labeled **Home Day**. This labeling is applied deterministically with no ambiguous cases. The resulting dataset comprises 222 Home Days (68.9%) and 100 University Days (31.1%).

3. Methodology

3.1 Data Preparation

Raw CSV data was loaded using the pandas library. The Date column was parsed into datetime format, and the dataset was sorted chronologically. Derived columns added include: Month, Month Name, Day-of-Week, Day-of-Week index (0 = Monday), Week-of-Year, Is-Weekend flag, and Year.

3.2 Descriptive Statistics

Standard descriptive statistics were computed for the full dataset and each day-type subset: mean, median, standard deviation, minimum, maximum, interquartile range, total sum, count, and coefficient of variation (CV = std / mean). Monthly breakdowns were generated using groupby aggregation.

3.3 Distribution Analysis

Histograms with kernel density estimation (KDE) were constructed using a bandwidth of 0.25. Bimodality was assessed visually and numerically. Overlapping KDE plots were used to compare Home and University Day distributions.

3.4 K-Means Behavioral State Modeling

A K-Means clustering algorithm (k = 3, n_init = 10, random_state = 42) was applied to the univariate kWh series to detect distinct behavioral regimes. The three clusters were post-hoc labeled Idle, Moderate, and Heavy based on their cluster centroid values in ascending order. Feature scaling was not applied because clustering was performed on a single univariate variable (daily kWh).

3.5 Time Series and Temporal Analysis

7-day and 14-day centered rolling averages were computed to smooth short-term variability. Rolling standard deviation (window = 14 days) was used as a proxy for behavioral variability over time. Calendar heatmaps were constructed with a Red-Yellow-Green diverging colormap to provide a visual overview of the entire year at a glance.

3.6 Power Draw Estimation

An approximate average wattage was estimated from the Home Day mean consumption under the assumption of 8 active computing hours per day: P = (kWh / 8 h) × 1000 W/kW. This estimate is acknowledged as a lower bound on peak instantaneous draw and an upper bound on the true average, as actual usage patterns include idle periods.

4. Results

4.1 Descriptive Statistics

Table 2 presents descriptive statistics for the full dataset and the two day-type subgroups. The analysis shows a large difference between Home and University days: 1.958 kWh vs. 0.595 kWh, a difference of 1.363 kWh per day and a 229% relative increase for home computing. The coefficient of variation (CV) for Home Days (21.5%) is notably lower than for University Days (41.8%), reflecting the more consistent nature of home computing routines.

Table 2. Descriptive statistics by day type.

Metric	All Days	Home Days	University Days
N (days)	322	222	100
Mean (kWh)	1.535	1.958	0.595
Median (kWh)	1.810	1.955	0.540
Std Dev (kWh)	0.734	0.420	0.249
Minimum (kWh)	0.19	0.20	0.19
Maximum (kWh)	3.61	3.61	2.31
IQR (kWh)	1.415	0.428	0.205
Total (kWh)	494.10	434.61	59.49
CV (%)	47.8%	21.5%	41.8%

An estimated average active power draw of approximately 245 W is derived from the Home Day mean under an 8-hour active usage assumption (1.958 kWh / 8 h ≈ 245 W). The peak estimated draw reaches ~451 W on the highest-consumption day (29 June 2025, 3.61 kWh).

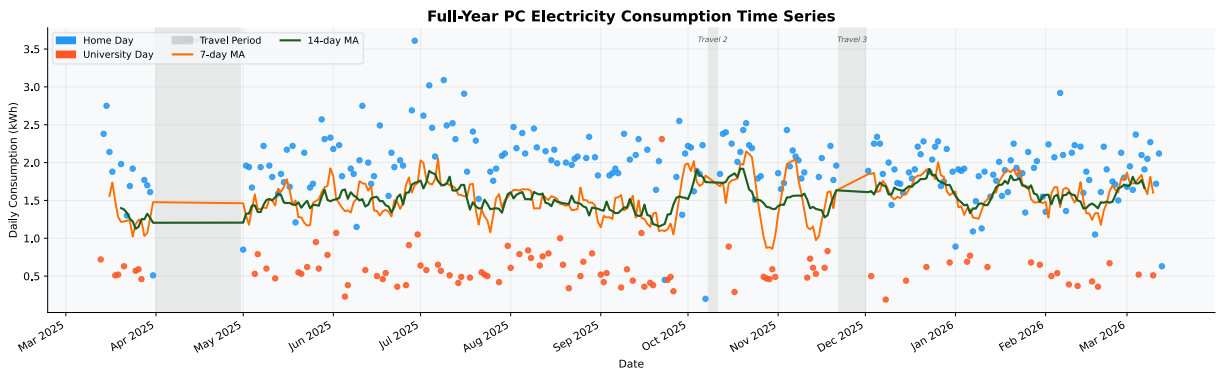

Figure 1. *Full-year time series of daily PC electricity consumption. Blue points = Home Days; orange points = University Days.*
Amber line = 7-day moving average; dark green line = 14-day moving average. Grey shaded regions indicate travel periods.

4.2 Monthly Analysis

Table 3 provides a month-by-month breakdown. December 2025 records the highest monthly total (51.49 kWh) despite the presence of a late-year travel interval, largely due to minimal university attendance (5 days). September 2025 has the lowest mean daily consumption (1.300 kWh/day), reflecting the highest university attendance frequency in the dataset (14 of 30 days).

Table 3. Monthly energy consumption summary.

Month	Total kWh	Mean kWh	Days	Home	Univ
Mar 2025 (partial)	25.63	1.349	19	12	7
Apr 2025	*- no observations-*				
May 2025	45.96	1.483	31	21	10
Jun 2025	46.34	1.545	30	19	11
Jul 2025	48.59	1.567	31	18	13
Aug 2025	47.72	1.539	31	18	13
Sep 2025	39.00	1.300	30	16	14
Oct 2025	43.41	1.608	27	20	7
Nov 2025	33.16	1.579	21	15	6
Dec 2025	51.49	1.716	30	25	5
Jan 2026	48.54	1.566	31	26	5
Feb 2026	42.79	1.528	28	21	7
Mar 2026 (partial)	21.47	1.652	13	11	2

Figure 3 shows the monthly trend of mean daily PC electricity consumption for the overall dataset together with the Home Day and University Day averages. Across the year, the Home Day curve remains consistently high while the University Day curve remains low, reinforcing the clear behavioral separation observed in the distribution analysis. The overall monthly average lies between the two and shifts upward or downward depending on the relative frequency of Home and University days in each month.

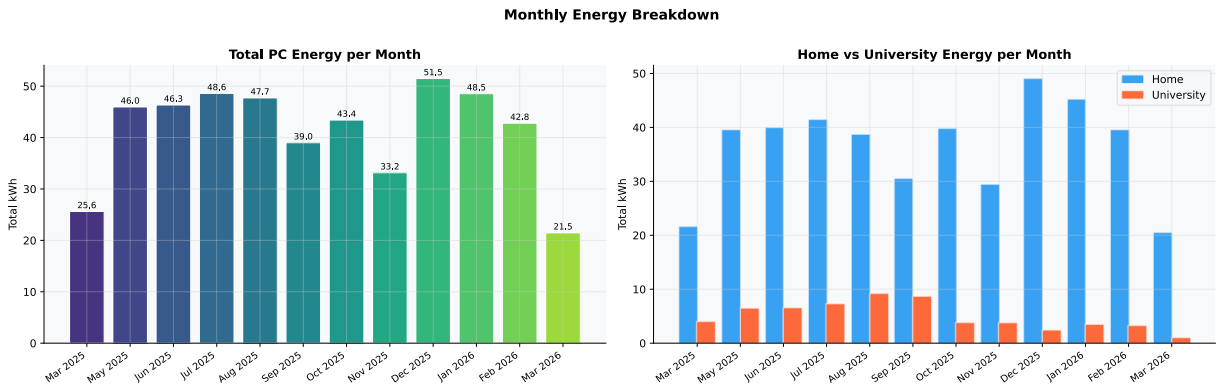

Figure 2. *Monthly PC electricity consumption totals (left) and monthly breakdown by Home and University days (right).*

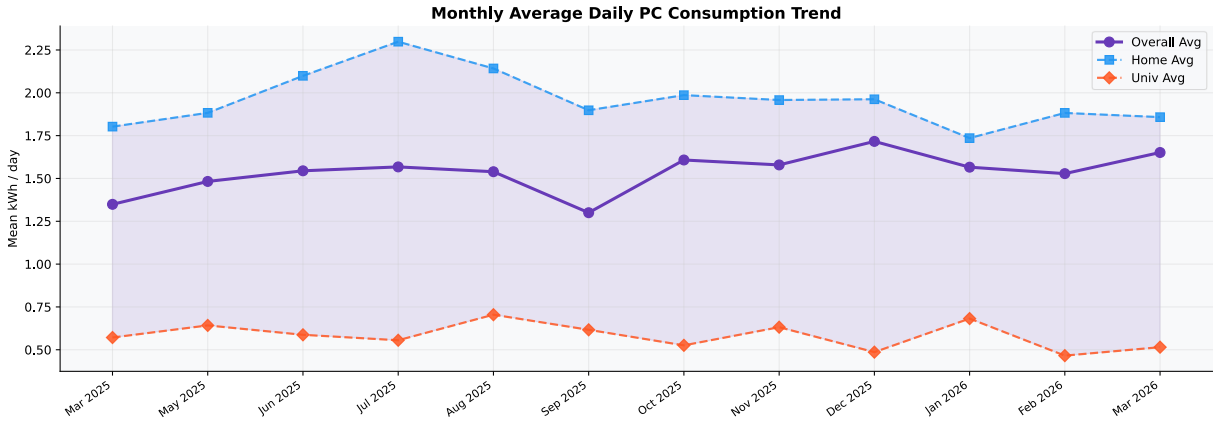

Figure 3. *Monthly trend of mean daily PC electricity consumption by month.*

4.3 Distribution Analysis

Figure 4 presents histograms and KDE curves for the full dataset and each day-type subset. The combined distribution is strongly bimodal, with one mode near 0.55 kWh (University/idle state) and a second broader mode near 1.80–2.00 kWh (Home/active state). Independently, each subgroup is approximately unimodal and near-symmetric. Home Days span roughly 1.5–2.5 kWh with minimal spread; University Days cluster tightly between 0.2–0.8 kWh. The overlap between the two distributions is negligible (Figure 5).

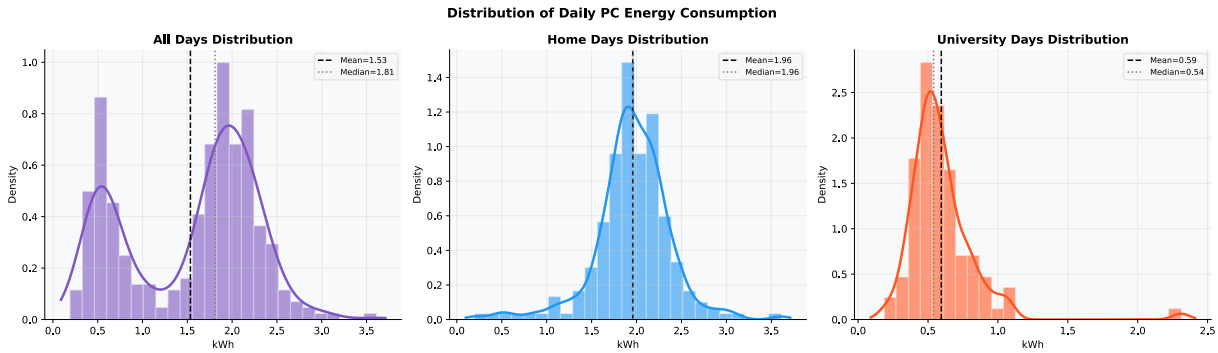

Figure 4. *Histograms and KDE curves for all days (left), Home Days (centre), and University Days (right). Dashed line = mean; dotted line = median.*

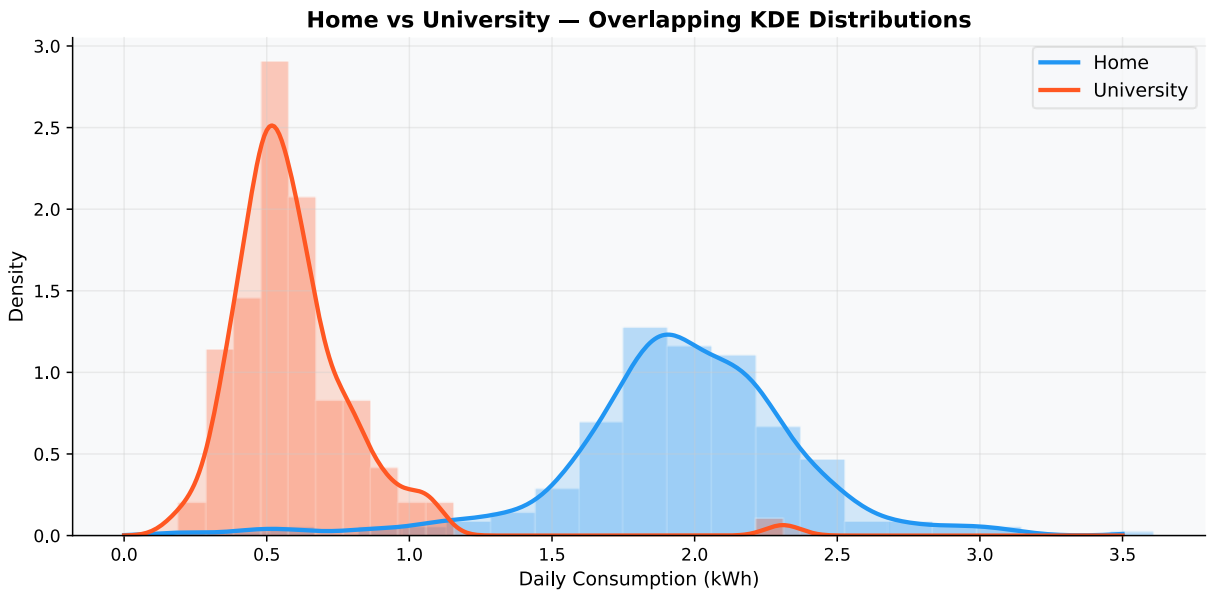

Figure 5. *Overlapping KDE distributions for Home Days (blue) and University Days (orange), demonstrating near-complete bimodal separation.*

4.4 Home vs University Comparison

Box plots and violin plots (Figure 6) confirm the visual impression from the KDE analysis. The notched box plot indicates that the difference in medians is statistically significant. The Home violin is compact and centered around ~1.96 kWh; the University violin is narrow and low, centered around ~0.54 kWh. All ten of the highest-consumption days occur on Home Days, while nine of the ten lowest occur on University Days.

Figure 7 shows the individual daily observations using a strip plot, illustrating the clear separation between Home and University Day consumption values and confirming that the two behavioral regimes exhibit minimal overlap.

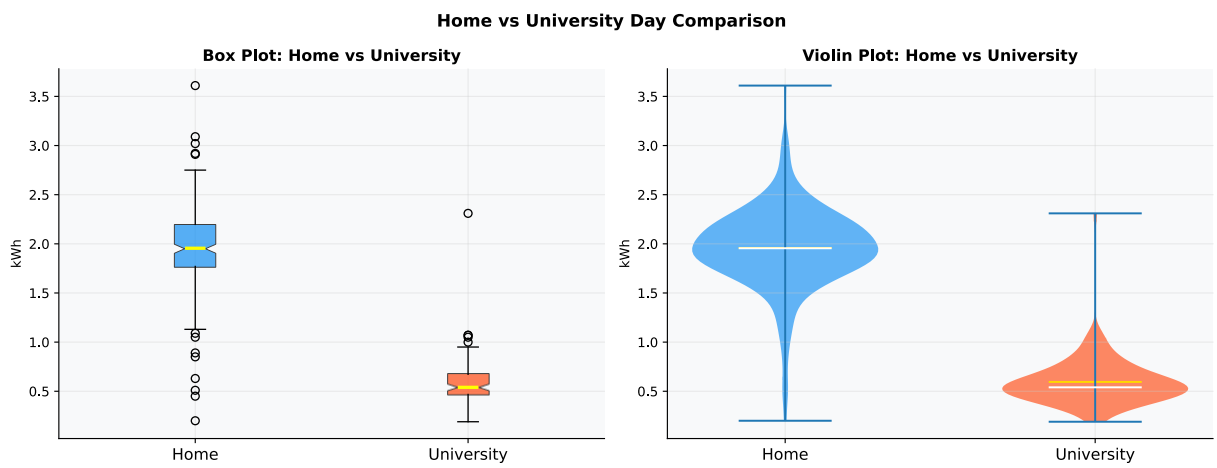

Figure 6. *Comparison of daily PC electricity consumption between Home and University days using box plots (left) and violin plots (right).*

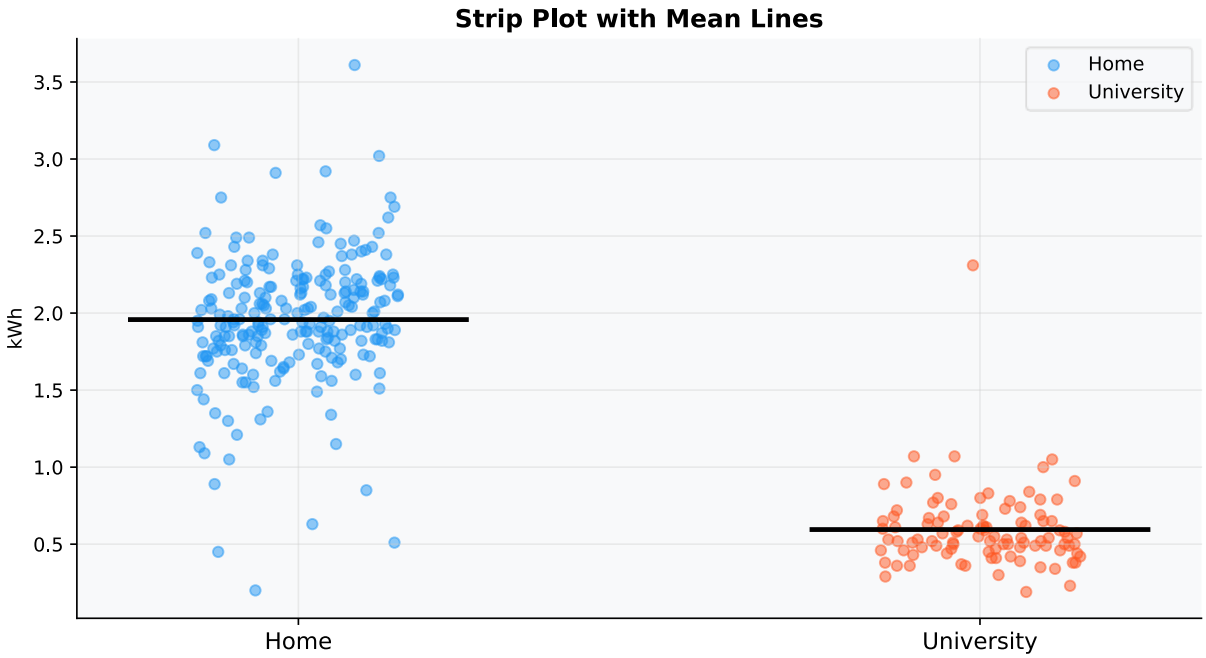

Figure 7. *Strip plot of daily PC electricity consumption for Home and University days with mean values indicated by horizontal bars.*

Table 4. Top 10 highest- and lowest-consumption days.

#	Date	kWh	Type	Day
1	29 Jun 2025	3.61	Home	Sunday
2	09 Jul 2025	3.09	Home	Wednesday
3	04 Jul 2025	3.02	Home	Friday
4	06 Feb 2026	2.92	Home	Friday
5	16 Jul 2025	2.91	Home	Wednesday
6	15 Mar 2025	2.75	Home	Saturday
7	11 Jun 2025	2.75	Home	Wednesday
8	28 Jun 2025	2.69	Home	Saturday
9	02 Jul 2025	2.62	Home	Wednesday
10	28 May 2025	2.57	Home	Wednesday

#	Date	kWh	Type	Day
1	08 Dec 2025	0.19	University	Monday
2	07 Oct 2025	0.20	Home	Tuesday
3	05 Jun 2025	0.23	University	Thursday
4	17 Oct 2025	0.29	University	Friday
5	26 Sep 2025	0.30	University	Friday
6	21 Aug 2025	0.34	University	Thursday
7	08 Sep 2025	0.35	University	Monday
8	23 Jun 2025	0.36	University	Monday
9	16 Sep 2025	0.36	University	Tuesday
10	19 Feb 2026	0.36	University	Thursday

The single anomalous Home Day in the low-10 list (07 October 2025, 0.20 kWh) occurs immediately before the second travel interval, suggesting the system was used only briefly before departure.

4.5 Calendar Heatmap

The month-by-month calendar heatmap (Figure 8) provides an overview of the entire year. The April 2025 void is particularly conspicuous. University visit days appear as cool green squares embedded in fields of warm yellow-red home-day squares throughout the rest of the year.

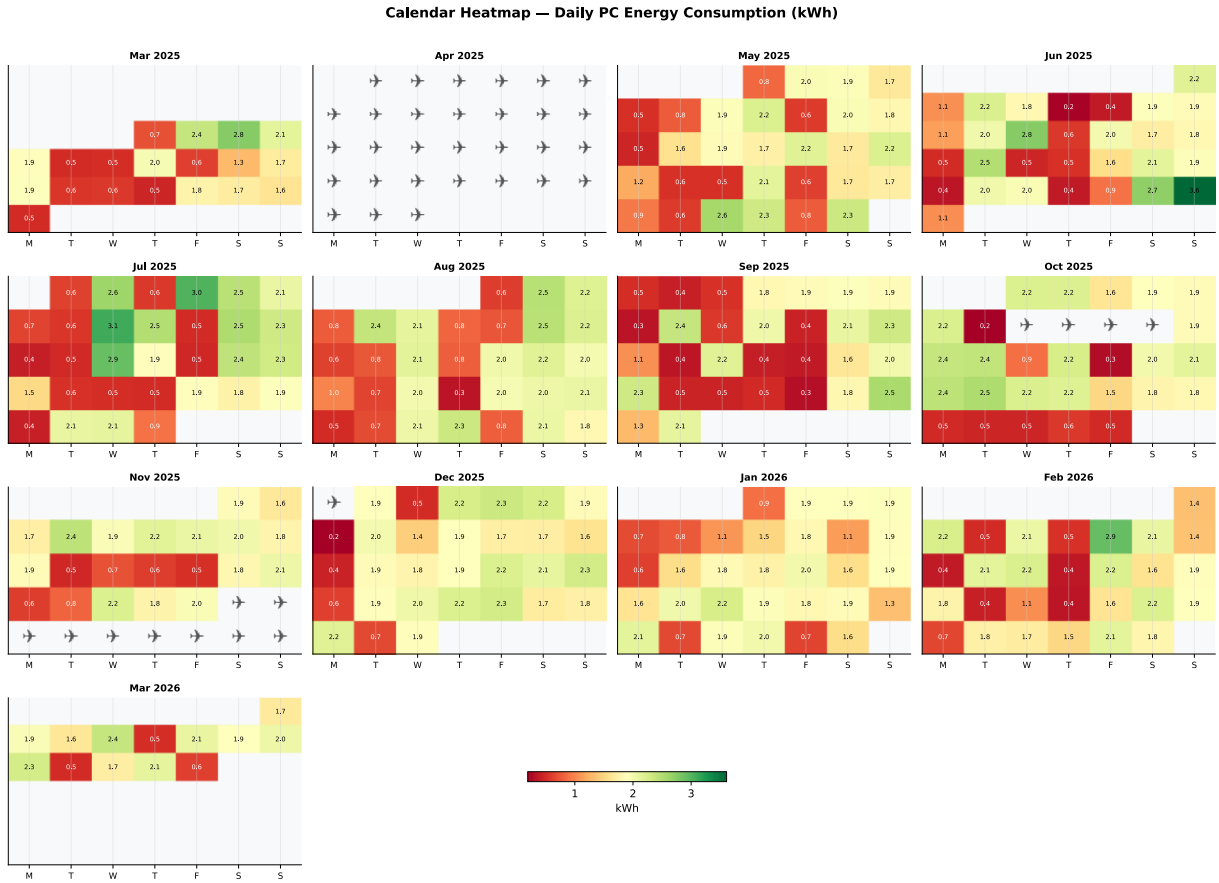

Figure 8. *Calendar heatmap of daily PC energy consumption. Green = low; red = high. Airplane symbol (✈) indicates days absent due to travel.*

4.6 Weekly Patterns

Box plots by day-of-week (Figure 9) reveal that university visits occur exclusively on weekdays, pulling weekday means downward. As a result, weekday and weekend means are closer than might be expected - weekdays include both high home-computing days and low university-attendance days, while weekends are almost entirely high-consumption home days. The day-of-week vs. month heatmap (Figure 10) highlights September–October 2025 as the lowest-consumption period across all weekdays. The weekday versus weekend comparison further illustrates this pattern, with weekends dominated by high-consumption Home Days and weekdays containing a mixture of high and low consumption depending on university attendance (Figure 11). Appendix Figure 16 shows the monthly counts of Home and University days, illustrating how variations in attendance frequency influence the monthly consumption patterns observed earlier. Appendix Figure 17 presents a scatter distribution of daily kWh by day-of-week, confirming that low-consumption observations are concentrated on weekdays corresponding to university attendance.

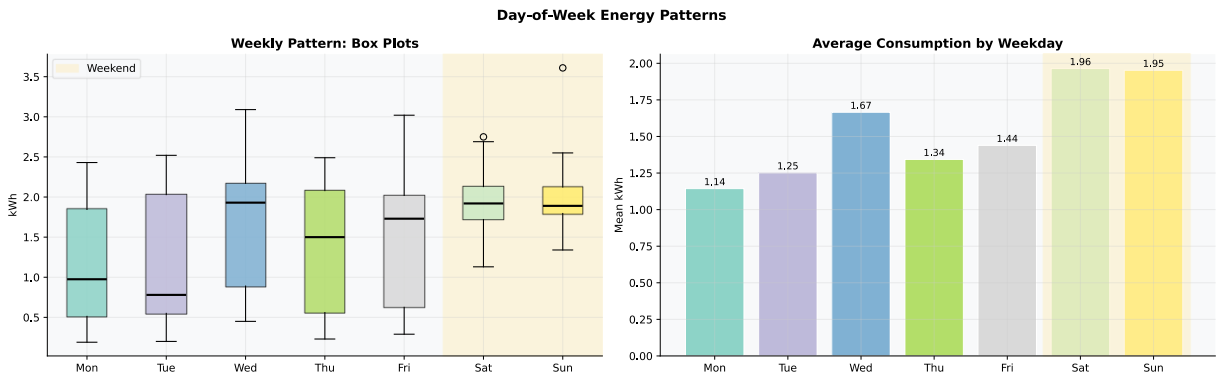

Figure 9. *Daily PC electricity consumption by day of week shown as box plots (left) and mean consumption bar chart (right). The shaded region indicates weekend days.*

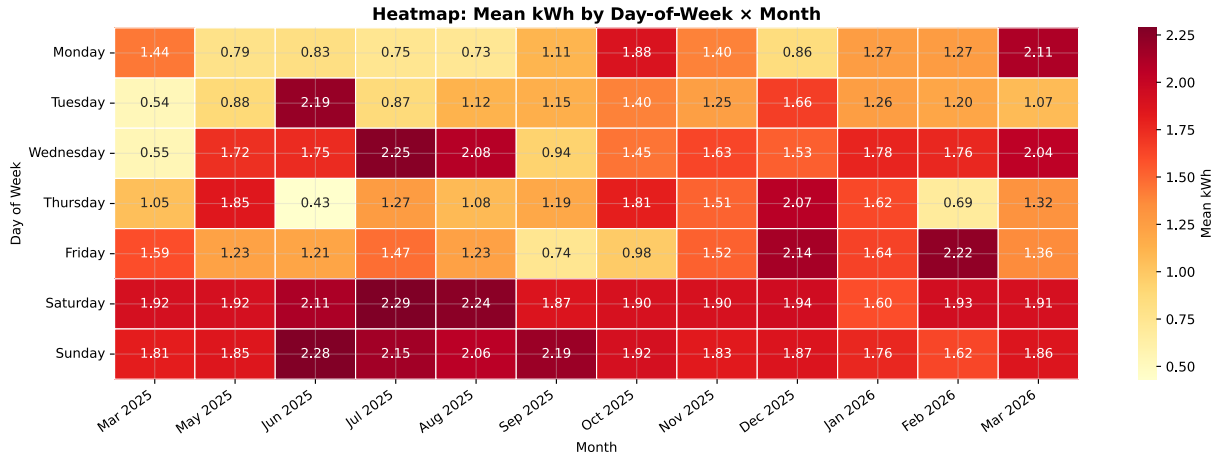

Figure 10. *Heatmap of mean daily kWh by day-of-week (rows) and month (columns). Lighter = lower consumption.*

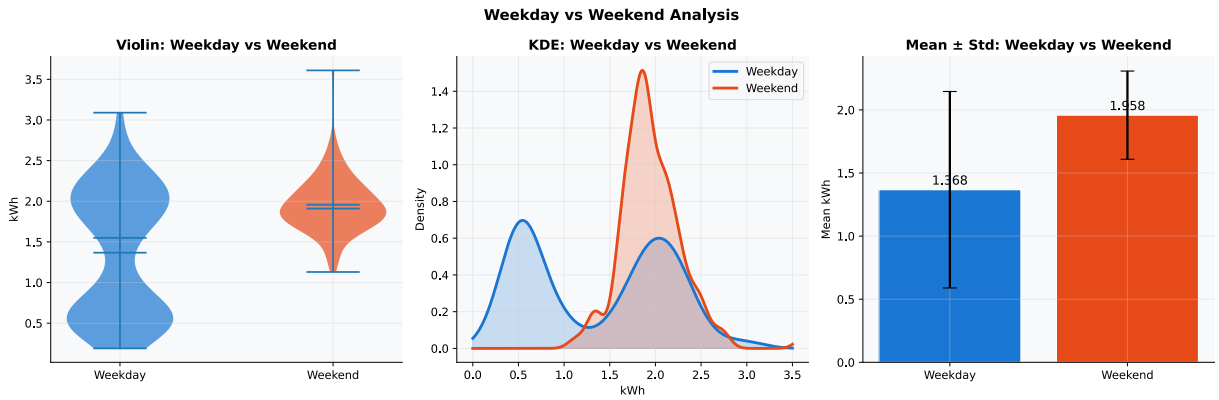

Figure 11. *Weekday vs weekend comparison of daily PC electricity consumption: violin plots (left), KDE distributions (center), and mean ± standard deviation bar chart (right).*

4.7 Behavioral State Clustering

K-Means clustering with k = 3 identifies three distinct usage regimes (Figure 12). Cluster centroids fall at 0.598 kWh (Idle, 109 days, 33.9%), 1.802 kWh (Moderate, 126 days, 39.1%), and 2.321 kWh (Heavy, 87 days, 27.0%). The Idle cluster is nearly co-extensive with the University Day set. The Moderate and Heavy clusters are both overwhelmingly composed of Home Days, with the Heavy cluster concentrated in the June–July 2025 period.

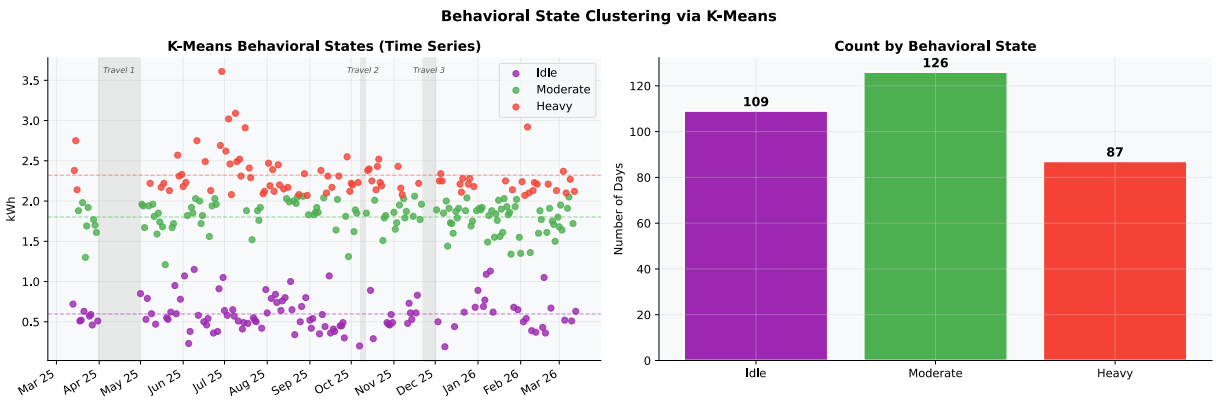

Figure 12. *K-Means behavioral state classification over the full year. Purple = Idle; green = Moderate; red = Heavy. Dashed lines indicate cluster centroids.*

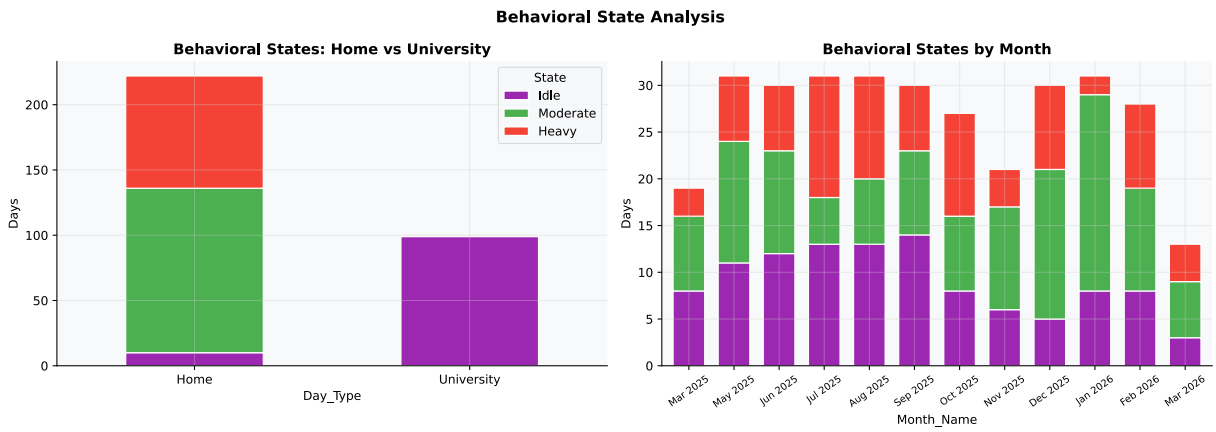

Figure 13. *Distribution of behavioral states (Idle, Moderate, Heavy) by day type (Home vs University).*

4.8 Cumulative Consumption and Temporal Variability

The cumulative energy curve (Figure 14) rises steadily throughout the year at an average rate of approximately 1.53 kWh/day, reaching 494.1 kWh by 13 March 2026. The three travel gaps appear as brief flat segments. The first travel interval forms the most prominent plateau, spanning the entirety of April 2025 and representing approximately 60 kWh of consumption that did not occur. Rolling variability analysis (Figure 15) shows that behavioral consistency is highest in the summer months (low rolling std) and highest variability occurs in the October–November transition period.

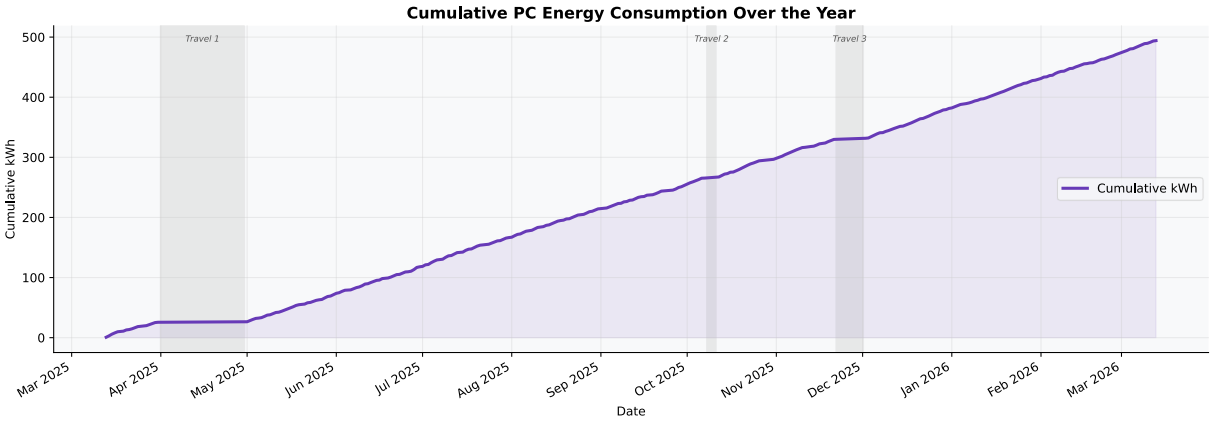

Figure 14. *Cumulative total PC energy consumption across the year. Grey shading indicates travel gaps.*

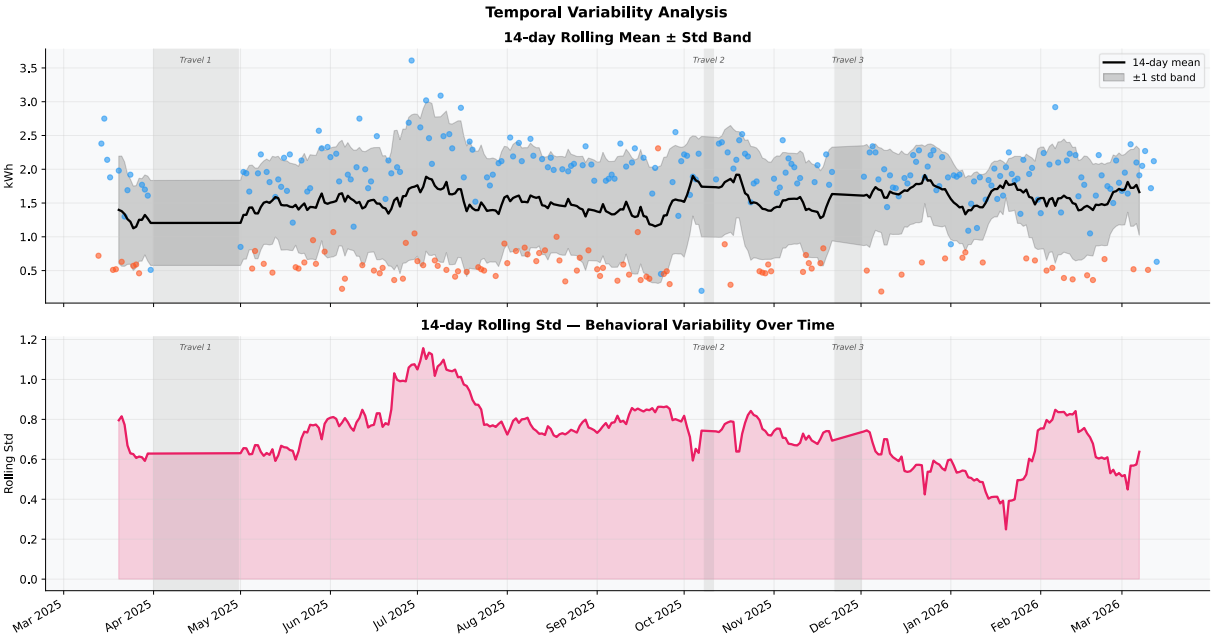

Figure 15. *Top: 14-day rolling mean with ±1 standard deviation band. Bottom: 14-day rolling standard deviation as a measure of behavioral variability.*

5. Discussion

5.1 The Smart Plug as a Behavioral Sensor

The near-complete bimodal separation between Home and University Day distributions implies that the electricity usage patterns can reveal behavioral activity patterns with very high accuracy - potentially exceeding 95% using a simple threshold classifier near ~0.9 kWh. This is a striking result for such a simple instrument. It suggests that, for a user with a consistent daily computing routine, a single smart plug energy monitor provides richer behavioral information than is typically appreciated. Crucially, this behavioral signal does not require the computing activity to be professional or research-oriented in nature; it arises simply from the presence or absence of sustained, active PC use over the course of a day. The smart plug responds to total energy drawn, regardless of whether the session

involves data analysis, creative work, gaming, or AI-assisted workflows - what matters is that the user is home and using the machine.

5.2 Habitual Computing Patterns

The coefficient of variation for Home Days (21.5%) is remarkably low for a year-long behavioral dataset. It would be a misreading of this consistency to attribute it exclusively to structured professional or research work. In practice, home computing on these days encompasses a broad range of activities - hobby projects, gaming, AI image generation and workflow experimentation, creative computing, and occasional research tasks - rather than a narrowly defined work routine. The finding is arguably more interesting for precisely this reason: it demonstrates that sustained, habitual computing - regardless of whether the activity is professional or recreational - produces a highly stable and reproducible daily energy signature. The person at the desk, engaged in what they find meaningful or enjoyable, draws power in a pattern that is as consistent across seasons and months as any formal work schedule. This suggests that behavioral regularity, not task type, is the primary determinant of the observed energy stability.

5.3 University Day Usage Pattern

The ~0.595 kWh mean for University Days does not reflect a system left idle or in standby. On these days the PC was switched off at the wall before leaving and powered on deliberately upon return. The usage pattern is better characterized as two short active sessions: a brief morning session immediately after waking, before commuting, and a second session in the evening after returning home - provided fatigue did not prevent it. Combined, these sessions total approximately 2–3 hours of active PC use per university day. This framing recontextualizes the 0.595 kWh figure: rather than representing a passive baseline or phantom load, it represents genuinely compressed but intentional computing activity, constrained by the practical demands of a commute day. The 100 University Days therefore collectively consumed 59.5 kWh - 12% of the annual total - not from standby draw, but from real, if abbreviated, sessions of use.

5.4 Post-Travel Reengagement Peak

The highest single-day consumption (3.61 kWh, 29 June 2025) occurs in the week immediately following the extended research travel interval. A cluster of high-consumption days in late June and early July 2025 is consistent with an intensive reengagement with home computing after a 30-day absence. After a month away from the machine, there is a natural accumulation of deferred activity - projects left mid-way, downloaded content to process, hobby workflows to resume, and a general re-immersion in the home computing environment. The elevated consumption in this period likely reflects longer-than-average sessions driven by this backlog, across the full mix of activities that characterize home use, rather than any single category of task.

5.5 Seasonal and Structural Trends

December 2025 and January 2026 record the highest monthly totals, not due to increased per-session intensity but due to the near-absence of university visits (5 days each month). With fewer commute days interrupting the routine, the home computing schedule becomes more continuous and unbroken. The late-year period (December–February) correspondingly shows the smallest gap between home and university day counts across the dataset. This pattern reflects reduced institutional obligations

rather than any change in the nature or intensity of home computing activity itself - the behavior at home remains consistent; it is simply given more days in which to occur.

6. Conclusion

This study analyzed one year of daily electricity consumption recorded from a smart plug connected to a personal desktop computer workstation and linked the energy data with a university attendance log. The results reveal a strong behavioral signal connecting physical work location and computing energy use. Home Days averaged 1.958 kWh per day compared with 0.595 kWh on University Days, representing a 229% difference in daily consumption. The overall distribution of electricity use is strongly bimodal, reflecting the two behavioral contexts. K-Means clustering further identified three usage regimes-Idle (~0.60 kWh), Moderate (~1.80 kWh), and Heavy (~2.32 kWh)-which align closely with the observed patterns of short commute-day sessions and extended home computing. Over the full observation period, 494.1 kWh were consumed across 322 recorded days, with 88% associated with home computing activity and 12% with shorter sessions on university-attendance days. These results show that even simple device-level energy monitoring can reveal consistent behavioral patterns in personal computing activity. Future work could extend this approach using higher temporal resolution data, monitoring additional devices, or integrating application-level usage logs to better characterize digital workload and computing behavior.

7. References

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A. Appendix - Additional Figures

This appendix contains all remaining visualizations generated during the analysis.

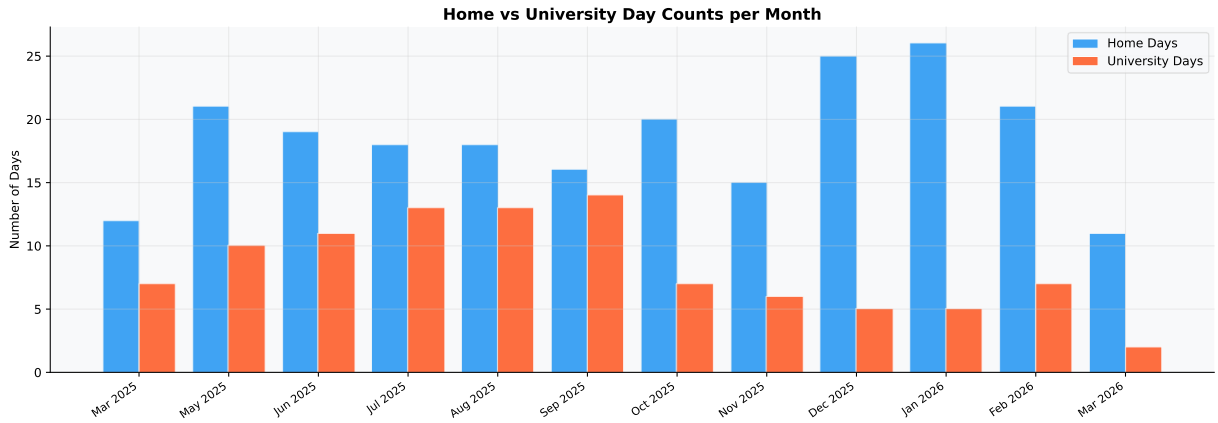

Figure 16. *Home vs University day counts per month.*

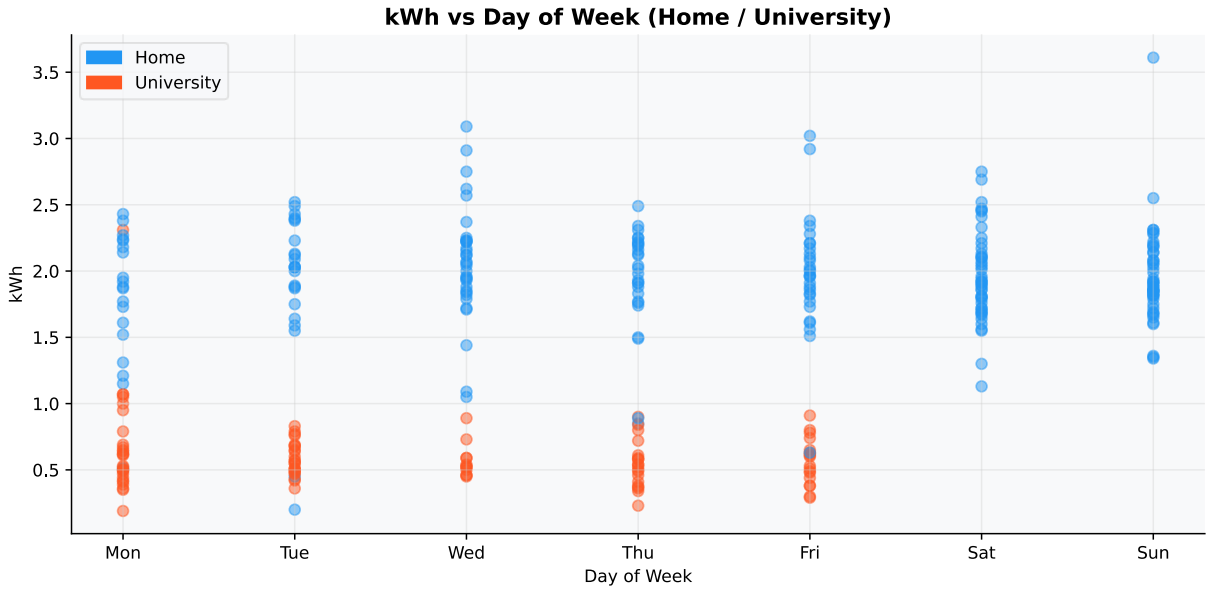

Figure 17. *Scatter plot of daily kWh by day-of-week, colored by Home (blue) / University (orange).*